LAB 4

CIRCUIT TECHNOLOGY

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1. Harmonic Shapers

Diode-based harmonic shaper circuit is used to convert a triangular input wave into a sine wave. A triangular wave can be obtained by integrating a rectangular (square) multi-vibrator signal

2. Optimal Oscilloscope Settings (Question 3)

The goal was to visually find optimal setting for potentiometers R10 and R11 using an oscilloscope. The function generator was set up to output a triangular wave with a frequency of 1000 Hz and an amplitude of 3V. By monitoring the signal on the oscilloscope, the potentiometers were adjusted simultaneously to shape the sharp triangular wave into a smooth sine wave. The optimal setting for both R10 and R11 was found to be **80%**.

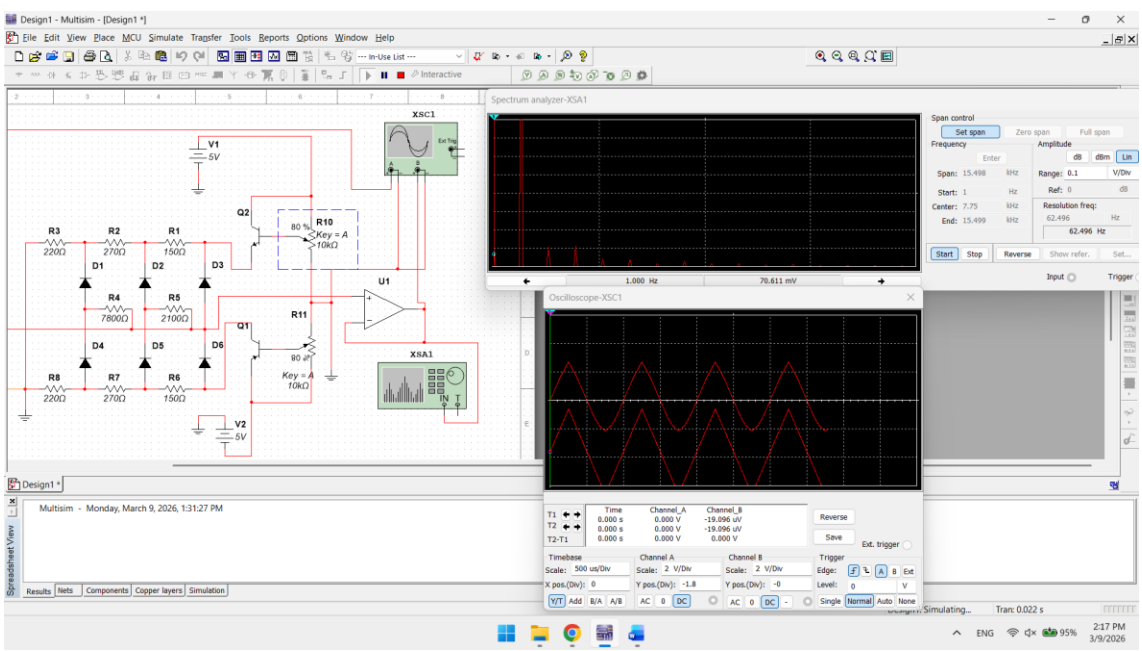

Figure 1. Schematic of the circuit, Oscilloscope, and Spectrum Analyzer

3. Total Harmonic Distortion (THD) Analysis (Question 4)

The objective was to tune R10 and R11 to obtain the minimum amount of higher harmonic components using a spectrum analyzer, and calculate the THD up to a maximum of 15 kHz.

Frequency (kHz)	Amplitude (V)
1 (primary f)	**2.138**
2	0.103223
3	0.10993
4	0.040565
5	0.035385
6	0.015659
7	0.026343
8	0.014268
9	0.016006
10	0.007392
11	0.010408
12	0.00252
13	0.004654
14	0.004899
15	0.002011

Table 1. Measured Amplitude V for frequencies (2kHz – 15kHz) at optimal 80% setting

The THD was calculated using the formula:

$$THD = \frac{\sqrt{\sum_{i=2}^{\infty} U_i^2}}{U_1}$$

$$\sum_{i=2}^{15} U_i^2 \; 0.010655 + 0.012085 + 0.001646 + 0.001252 + 0.000245 + 0.000694 + 0.000204 + 0.000256 + 0.000055 + 0.000108 + 0.000006 + 0.000022 + 0.000024 + 0.000004 = 0.027256\, V^2$$

$$\sqrt{0.027256} = 0.16509\, V$$

$$THD = \frac{0.16509}{2.138} = 0.0772 = \mathbf{7.72}\%$$